National University of Computer and Emerging Sciences

Karachi Campus

MID IISolution-Artificial Intelligence (AI-2002)

Question 01 (1 mark each)

1. Ordering decides the sequence of decisions (0.25). Filtering removes invalid options (0.25). Forward Checking "Look ahead" filtering to avoid future conflicts. (0.5)
2. **Strengths:** Fast, simple, works well for text and high-dimensional data (0.4)
   Weakness: Unrealistic independence assumption + zero probability issue (0.4)
   Solution: *Feature Selection / Bayesian Networks* → mitigates independence assumption (0.2)
3.

Aspect	Mini-Max	Alpha-Beta Pruning	
Worst Time	(O(b^d))	(O(b^d))	(0.2 marks)
Best Time	(O(b^d))	(O(b^{d/2}))	(0.4 marks)
Space	(O(b.d))	(O(b.d))	(0.4 marks)

Question 02: Part (a)

1. Yes = 6, No = 6 → P(Yes) = P(No) = 6/12 = 0.5; Yes, class (rows 1, 3, 4, 7, 9, 11 — six total): (1 mark)

- Age=Old: rows 1, 3, 7, 9, 11 → 5/6 = 0.833
- Cholesterol=High: rows 1, 3, 4, 7, 11 → 4/6 = 0.667 (row 9 is Low)

2. No class (rows 2, 5, 6, 8, 10, 12 — six total): (1 mark)

- Age=Old: rows 5, 12 → 2/6 = 0.333
- Cholesterol=High: row 8 only → 1/6 = 0.167

3. Posteriors for new patient (Old, High): (1 mark)

- P(Yes | Old, High) ∝ 0.5 × 0.833 × 0.667 = 0.278
- P(No | Old, High) ∝ 0.5 × 0.333 × 0.167 = 0.028

Since 0.278 > 0.028 → Heart Disease = Yes

Question 02: Part (b)

a) P(C)·P(L|C)·P(¬M|C)·P(¬B|L,M) (1.5 marks)
   • $P(C = true)$ = 0.1
   • $P(L = true \mid C = true)$ = 0.2
   • $P(M = true \mid C = true)$ = 0.4⇒ $P(\neg M \mid C)$ = 0.6
   • $P(B = true \mid L = true, M = false)$ = 0.9⇒ $P(\neg B)$ = 0.1

   Answer: 0.0012

b) Enumeration: Needs all combinations → slow (0.5 marks)
   Bayesian Network: Uses independence and Fewer computations
   Conclusion: Bayesian networks reduce exponential complexity to efficient factorized computation

Question 02: Part (c) (0.5 each)

1)
$$P(\text{Graduate} \mid \text{Online}) = \frac{P(\text{Graduate} \cap \text{Online})}{P(\text{Online})}$$
$$= \frac{0.12}{0.35} = 0.3429$$

2)
$$P(\text{Hybrid} \mid \text{Undergraduate}) = \frac{0.10}{0.50}$$
$$= 0.20$$

3)
First compute:
$$P(\text{PhD} \mid \text{OnCampus}) = \frac{0.03}{0.45} = 0.0667$$
So,
$$P(\text{Not PhD} \mid \text{OnCampus}) = 1 - 0.0667 = 0.9333$$

4)
First find:
$$P(G \cup PhD) = 0.40 + 0.10 = 0.50$$
Online among these groups:
$$P(O \cap (G \cup PhD)) = 0.12 + 0.05 = 0.17$$
Now:
$$P(O \mid G \cup PhD) = \frac{0.17}{0.50} = 0.34$$

Question 3: Part (a)

Variables: 0.25
Domains: 0.25
Constraints: 0.5
Scheduling: 1.25

Constraints:

- 3 Days × 3 Shifts = **9 total assignments**
- Nurses: **N1, N2, N3, N4**
- Each shift → **exactly 1 nurse**
- Each nurse → **≤ 1 shift per day**
- **N1 ✕ Night shifts**
- **N2 ≥ 2 Morning shifts**
- **N3 & N4 ✕ same shift on same day**

Day	Morning (M)	Evening (E)	Night (N)
Day 1	N2	N3	N4
Day 2	N2	N1	N3
Day 3	N3	N4	N2

b) Yes, multiple solutions exist. (0.25)

Justification:

- This is a **Constraint Satisfaction Problem (CSP)** with:
  - Multiple variables (9 shifts)
  - Flexible domains (4 nurses)
- Constraints **restrict but do not fully determine** assignments
- Different valid permutations exist:
  - Swap N3/N4 across days (while respecting constraint)
  - Reassign N2's required morning shifts across different days
  - Use N1 in different non-night slots

Hence, the solution space is **not unique**, and **many valid schedules** can satisfy all constraints.

Question 03: Part (b)

Pruned Branches shown properly: 1
Values shown at node properly: 1
Traversing shown properly: 0.5